

BirdGuard - Design Analysis

Design 1

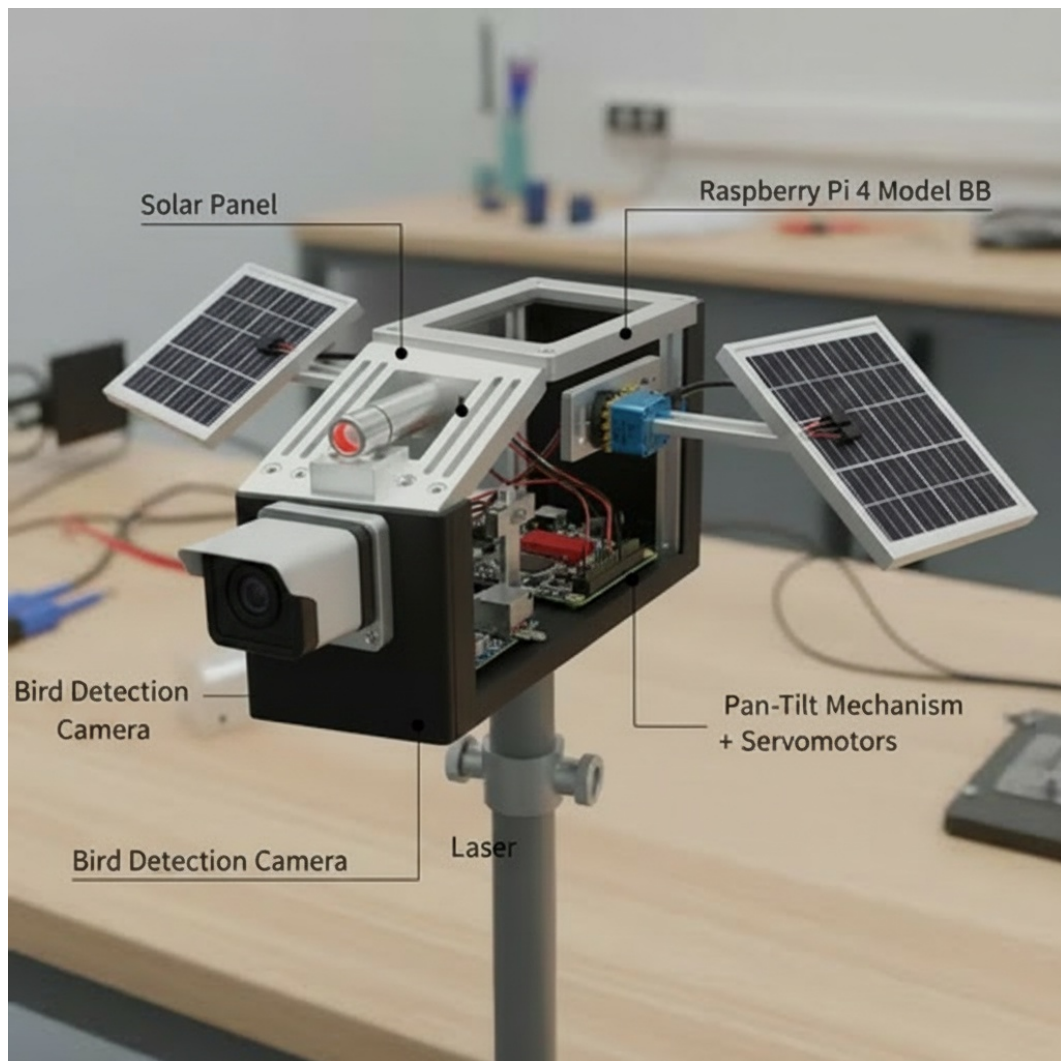


Figure 1: Labeled CAD render showing internal components and structure

Pros

- Aluminum extrusion frame provides a rigid, modular structure that is easy to modify or expand.
- Dual solar panels positioned on both sides give balanced power generation across different sun angles.
- Laser and pan-tilt mechanism are centrally mounted, keeping the centre of gravity low and stable.
- Raspberry Pi 4 is enclosed within the frame, protecting it from weather while remaining accessible.
- Open-frame design allows good airflow around electronics, reducing thermal throttling risk.
- Camera is forward-facing and clearly separated from the laser, reducing lens flare interference.
- Compact footprint on the mounting pole suits rooftop or post installation in tight spaces.

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Cons

- Aluminum extrusion frame adds weight compared to a 3D-printed or sheet-metal alternative.
- Two solar panels increase material cost and complexity of the wiring harness.
- The exposed wiring visible in the render is a maintenance and reliability concern outdoors.
- Pan-tilt servo mounting appears cramped inside the frame, potentially limiting range of motion.
- No visible battery housing -- energy storage placement is unclear from this design.
- Single camera angle may create blind spots for birds approaching from the sides or rear.

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Design 2



Figure 2: Photorealistic render showing field deployment on a farm fence post

Pros

- Fence-post mounting is practical and widely available in farm environments with no extra infrastructure needed.
- Dual solar panels on a cross-bracket spread power generation and keep the unit self-sufficient off-grid.
- Weatherproof camera housing looks robust and suitable for outdoor all-season operation.
- BirdGuard branding on the unit presents a polished, product-ready aesthetic.
- Compact, self-contained unit on a single post is easy to reposition across a field.
- Clean cable management visible along the post reduces snagging and wear on wiring.

Cons

- Fence-post mount may not provide enough height for wide-area laser coverage in tall-crop fields.
- Solar panel cross-bracket may act as a wind sail, risking unit instability in high-wind rural areas.
- Single camera lens shown may not have wide enough field of view for full perimeter detection.
- Post mounting limits aiming flexibility -- a fixed post cannot track birds moving outside the laser arc.
- No visible weatherproof seal or IP rating indicator, making environmental protection unclear.
- Cable running down the post is exposed and could be damaged by animals or farm machinery.